1. Current State of Research in the Field

1.1 Basis and Problem Statement

Human collaboration remains central to knowledge-intensive work yet systematically underperforms due to well-known process losses such as production blocking, hidden profiles, dominance effects, and premature convergence, which erode effectiveness, efficiency, and satisfaction (Stasser & Titus, 1985; Nunamaker et al., 1991; Dennis & Valacich, 1993; Dennis, Wixom, & Vandenberg, 2001; Briggs, de Vreede, & Nunamaker, 2003). Group Support Systems (GSS) demonstrated substantial gains by enabling parallel input and anonymity and by embedding process structure; outcomes improved when facilitation aligned tool use with tasks (Nunamaker et al., 1991; Dennis, Wixom, & Vandenberg, 2001; Briggs, de Vreede, & Nunamaker, 2003). Facilitation, such as clarifying roles, balancing participation, structuring phases, and guiding consensus, mitigates process losses, but skilled facilitators are scarce and unevenly available. This scarcity motivates digital augmentation of facilitation across settings where collaboration quality is mission-critical (e.g., education, healthcare, public administration; see e.g., Dennis, Wixom, & Vandenberg, 2001; Briggs, de Vreede, & Nunamaker, 2003). Modern AI with multimodal perception and behaviour makes the augmentation of facilitation a design choice rather than a constraint, motivating a theory-driven extension of existing models (Dennis, Wixom, & Vandenberg, 2001; Seeber et al., 2020).

To assess practical demand, we conducted 50 interviews with academic and professional facilitators. Participants consistently anticipated near-term AI augmentation of their practice, alongside concerns about legitimacy and control. This raises three project-shaping questions: (1) Which facilitation capabilities can be delegated to AI? (2) Under what conditions does delegation improve or harm outcomes? (3) How should humans and AI co-orchestrate facilitation in hybrid teams?

1.2 Facilitation in Group Support Systems

The Fit-Appropriation Model (FAM) explains collaboration outcomes as a function of task-technology fit and appropriation, i.e. how groups actually use system features during work (Dennis, Wixom, & Vandenberg, 2001). In the GSS tradition, facilitation was pivotal to effective appropriation: outcomes depended on how well facilitators guided the use of features that enable parallelism, anonymity, and process structure (Nunamaker et al., 1991; Briggs, de Vreede, & Nunamaker, 2003). Yet FAM typically treated facilitation as an undifferentiated support, leaving unspecified which capabilities matter and under what conditions. We therefore adopt facilitation capabilities as the unit of analysis, distinguishing communication support (how contributions are exchanged: participation balancing, turn-taking, pacing, climate) from information-processing support (what happens to contributions: capture, structure, summarisation, evaluation). Either class can, in principle, be enacted by humans or digital artefacts; their allocation affects effectiveness, efficiency, and satisfaction (Nunamaker et al., 1991; Dennis, Wixom, & Vandenberg, 2001; Briggs, de Vreede, & Nunamaker, 2003). This capability view provides micro-foundations for FAM and yields testable predictions for hybrid facilitation.

1.3 Human-AI Collaboration

Research on human-AI teaming has progressed rapidly, often in command-and-control contexts (e.g., military operations, constrained multiplayer games) where roles and communication are tightly structured (Lyons et al., 2021; O'Neill et al., 2022). Much of this work focuses on dyads or micro-teams, limiting applicability to typical knowledge work with 4-8 participants and fluid interaction (Bankins et al., 2024; Burton et al., 2024; Fischer et al., 2024).

The rise of generative AI extends support of collaboration beyond narrow-scope automation. Systems can contribute ideas, provide feedback, synthesise input, and adopt roles (e.g., critic, creative partner, proxy), suggesting support across generate-evaluate-decide task families (Dennis, Wixom, & Vandenberg, 2001; Rai et al., 2019; Dennis et al., 2023). However, most applications remain individual- or assistant-centric, lacking the awareness and predictable structure required for multi-party collaboration (Maedche et al., 2019; Nah et al., 2023; see e.g., Kar et al., 2023; Riemer & Peter, 2024; Vats et al., 2024). Calls to embed AI directly into team decision-making have emphasised the need for systematic theory and validated design principles (Parker & Grote, 2022; Anthony et al., 2023; Bankins et al., 2024; Fischer et al., 2024; Iftikhar et al., 2024). Our project responds to this call by embedding AI into small-group, process-aware collaboration, with a focus on facilitation as the central mechanism.

1.4 Human-AI Teams and Co-Facilitation

Rather than automate facilitation end-to-end, we target AI co-facilitation: digital artefacts enact selected facilitation capabilities while a human facilitator retains legitimacy, norm sensitivity, and contextual judgement, consistent with the principles of Hybrid Intelligence (Dellermann et al., 2019; Li et al., 2024). AI often excels at consistency, timing, and scale (e.g., enforcing timeboxes, tracking participation, performing high-volume analysis), whereas humans excel at interpretation, fairness concerns, and authority calibrated to context (Hemmer et al., 2021; Lou et al., 2025). The central design problem is therefore allocation: which capabilities to delegate, which to retain, and how to coordinate them so that groups remain both efficient and legitimate (Seeber et al., 2020; Fragiadakis et al., 2024).

Computational social science demonstrates the feasibility of multimodal sensing (Lehmann-Willenbrock & Hung, 2024; Järvelä et al., 2025), while HCI/IS research highlights transparency, trust calibration, and human override as safeguards for responsible AI use (Shneiderman, 2020; Lepri et al., 2021; Xu et al., 2023). What is missing is a systematic account of which facilitation capabilities should be enacted by AI vs. humans, under what conditions, and with what effects—a gap explicitly noted in recent reviews of Hybrid Intelligence (Hemmer et al., 2021; Nguyen, 2025). This is the focus of our project.

1.5 Research Needs and Project Contributions

Three limitations emerge across the reviewed literatures. First, while FAM recognises facilitation, it treats it as an undifferentiated appropriation support; a capability-level micro-foundation is missing (Dennis, Wixom, & Vandenberg, 2001; Briggs, de Vreede, & Nunamaker, 2003). Second, advances in human-AI collaboration (incl. GenAI) have not been integrated into theories of group facilitation; most systems remain assistant-centric and dyadic (Maedche et al., 2019; Nah et al., 2023; Seeber et al., 2020). Third, progress in multimodal sensing and responsible AI lacks a principled mapping to which facilitation capabilities can/should be enacted by AI vs. humans, and when (transparency, override; see e.g., Rai et al., 2019).

We address these gaps by **(1) extending FAM** with capability-level constructs of communication and information-processing support and **(2) designing and evaluating** six multimodal co-facilitation artefacts that provide causal evidence on allocation effects (human vs. AI vs. hybrid; advisory vs. assertive) on effectiveness, efficiency, satisfaction, and legitimacy. This positions our work to derive predictive rules and design principles for hybrid facilitation, culminating in an extended FAM and a design theory of multimodal co-facilitation (Gregor & Jones, 2007; Gregor et al., 2020).

2. State of Own Research

This proposal combines UZH's expertise in collaborative technologies with ZHAW's technical depth designing Human-AI collaboration. This integration is essential to achieve the project's goal: creating digital co-facilitators that are theoretically grounded, technically robust, and experimentally validated.

Gerhard Schwabe, UZH

Prof. Gerhard Schwabe has researched technology-supported collaboration since the 1990s, starting with Group Support Systems (GSS) at the University of Arizona (Nunamaker et al., 1991; Schwabe, 1995). His work has since advanced collaboration engineering across domains such as education, city councils, tourism, financial advice, and healthcare (Schwabe & Göth, 2005; Giesbrecht et al., 2017; Staehelin et al., 2024a, 2024b). A recurring theme is the design and validation of reusable facilitation structures ('thinklets') for advice-giving and decision-making.

More recently, his group has pioneered human-AI collaboration in service and decision contexts, developing conversational agents as third partners in advice-giving (Eckhardt et al., 2023, 2024) and contributing to the widely cited "Machines as Teammates" research agenda (Seeber et al., 2020). Since the emergence of generative AI, they have prototyped and experimentally validated AI-supported systems for public services, crisis response, digital health, and finance, yielding design principles, architectural guidelines, and responsible-use frameworks (Dolata et al., 2022, 2023; Wu et al., 2024). In 2023, Prof. Schwabe initiated a collaboration with the GSS provider **XLeap** to explore AI integration into professional facilitation software. Early results include an AI brainstorming participant (Schwabe et al., 2025) and ongoing experiments with AI-supported selection and ranking. In parallel, he conducted 50 interviews with academic and professional facilitators, which identified co-facilitation as a key emerging practice and directly informed the hypotheses of this proposal.

This research trajectory links the origins of GSS to today's generative AI: from codifying human facilitation expertise into reusable process patterns, to designing AI agents that enact facilitation capabilities in hybrid teams. It positions Prof. Schwabe as a leading contributor to theory-driven, design-science research on collaboration systems and their evolution towards human-AI co-facilitation.

Alexandre de Spindler, ZHAW

Prof. Alexandre de Spindler brings over two decades of experience designing and evaluating interactive systems at the intersection of information systems, human-computer interaction (HCI), and human-AI collaboration. His early research pioneered frameworks for mobile and web-based interaction, including context-aware social applications (de Spindler et al., 2009a; de Spindler et al., 2010a), ambient wall displays (de Spindler et al., 2010b), and cooperative visualization environments (de Spindler et al., 2009b; de Spindler et al., 2009c). These contributions foregrounded user experience and participatory design. Since 2010, he has led multiple Innosuisse and EU AAL projects applying participatory design and usability evaluation to healthcare, creative workshops, and collaborative learning. Across these efforts, the common thread is translational HCI: turning stakeholder requirements into robust, adoptable systems validated in multi-site field studies.

Building on this HCI foundation, his recent work advances human-AI collaboration through the PROMISE (Wu et al., 2024; Heierli et al., 2024; Heierli et al., 2025) and Social Behaviour Regulation (SBR) (Komenda et al., 2025) frameworks. PROMISE orchestrates LLM behaviour via state machines to ensure process predictability and role separation. It has been validated in healthcare (adherence and health-literacy companions deployed in multi-week studies with patients and

physicians; (Färber et al., 2023; Wu et al., 2024; Meng et al., 2024) and in financial advisory pilots funded by Innosuisse, demonstrating reproducible information-processing support suited for co-facilitation artefacts. SBR extends PROMISE with multimodal sensing and socially adaptive interventions (verbal, para-verbal, and selected non-verbal cues), equipping agents with sensitivity to group dynamics (Komenda et al., 2025). It was first demonstrated in social learning and conflict-handling studies and is now being ported to VR-based training environments.

Together, PROMISE (predictability) and SBR (social adaptivity) form the technical backbone for WP2-WP4; both frameworks are peer-reviewed, open source, and validated across healthcare, education, finance, and workshops, positioning de Spindler's work as enabling advances in digital co-facilitation.

Alan Dennis, Indiana University

Prof. Alan Dennis has conducted research on technology-supported group decision-making since the late 1980s. His early work at the University of Arizona developed and empirically tested Group Support Systems (Nunamaker et al., 1991), leading to the formulation of the Fit-Appropriation Model (FAM) (Dennis, Wixom, & Vandenberg, 2001), which remains a central framework for explaining group outcomes in technology-supported collaboration. He also advanced the theory and practice of electronic brainstorming, showing that parallel digital communication can overcome production blocking and enable groups to outperform individuals (Dennis & Valacich, 1993; Valacich & Dennis, 1994; Dennis et al., 1996).

His research refined mechanisms such as synergy, problem decomposition, and consensus formation (Dennis et al., 1997; Bhagwatwar et al., 2013, 2018; Minas et al., 2018), providing the theoretical foundation for extending FAM with facilitation capabilities. In recent years, Prof. Dennis has expanded this trajectory to AI-based collaboration, examining conversational agents such as chatbots and digital humans (Seymour et al., 2020, 2021, 2023, 2025). Prompted by Schwabe and colleagues' "Machines as Teammates" agenda (Seeber et al., 2020), he investigated user trust and willingness to collaborate with AI team members (Dennis et al., 2023). This work provides essential groundwork for theorising hybrid facilitation and integrating digital agents as co-facilitators.

These contributions link the origins of GSS and FAM with current research on AI teammates, ensuring that this project's empirical findings are embedded in a coherent theoretical lineage and translated into a mid-range theory of human-AI co-facilitation.

3. Detailed Research Plan

The overall goal of this project is to design and evaluate digital co-facilitators that extend the Fit-Appropriation Model (FAM) by opening the "black box" of facilitation. We treat facilitation not as an undifferentiated appropriation factor but as a structured set of communication support and information-processing support capabilities. These capabilities can be enacted by humans or by digital artefacts (prototypes) with multimodal perception and behaviour, and their allocation directly affects group outcomes in terms of effectiveness, efficiency, and satisfaction. We design digital co-facilitators as human-centred AI artefacts that extend, not replace, human agency in group facilitation.

To achieve this goal, we pursue the following two interconnected objectives.

Theory & Evidence. Extend the Fit-Appropriation Model (FAM) with capability-level constructs for facilitation (communication and information-processing) and provide causal evidence by testing whether delegating these capabilities to AI improves group outcomes (effectiveness, efficiency, satisfaction, legitimacy) relative to human-only facilitation.

Design & Principles. Design and implement digital artefacts (prototypes) that instantiate selected facilitation capabilities using multimodal perception (verbal, para-verbal, non-verbal cues) and multimodal behaviour (spoken, textual, visual interventions), and use these artefacts as expository instantiations (Gregor & Jones, 2007) to refine and articulate a design theory of multimodal co-facilitation.

Beyond advancing Information Systems theory by extending FAM, the project establishes an explicit connection to foundational theories of Group Processes, Organizational Behaviour, and Social Psychology. Specifically, our capability-based extension of FAM is micro-founded on principles from Group Process Loss/Gain Theory (e.g., production blocking, hidden profiles; Stasser & Titus, 1985) which were later elaborated in the GSS tradition (Nunamaker et al., 1991; Dennis & Valacich, 1993), Organizational Justice Theory (Colquitt, 2001), and research on Shared Mental Models and Team Cognition (Mohammed et al., 2010). Core constructs studied in these fields, such as participation equality, trust (McAllister, 1995), legitimacy and confidence (Johnson et al., 2014), satisfaction (Hackman, 1974), and process clarity (Dennis et al., 1996), become the measurable outcomes of our capability allocation hypotheses. This integration allows us to generate predictive rules about hybrid co-facilitation that are theoretically grounded across disciplinary boundaries.

3.1 Conceptual Framing

Figure 1 summarises our framing. In the original FAM, facilitation is treated as a generic appropriation support. We extend this by distinguishing two classes of facilitation capabilities: communication support (e.g., regulating participation, managing tone, pacing activities) and information-processing support (e.g., summarising, clustering, testing consensus). Each capability can be enacted by a human facilitator, by a digital artefact, or in hybrid combinations (Dennis, Wixom, & Vandenberg, 2001; Nunamaker et al., 1991; Briggs, de Vreede, & Nunamaker, 2003).

Modern AI systems enable multimodal perception and behaviour, capturing verbal (words, semantic structures), para-verbal (tone, pauses, interruptions, laughter), and non-verbal (gaze, gestures, facial expressions) cues, and responding through spoken, textual, or visual interventions. This makes the allocation of facilitation capabilities between humans and digital artefacts a designable choice rather than a technological constraint (Seeber et al., 2020; Dellermann et al., 2019).

The group outcomes (effectiveness, efficiency, and satisfaction) depend on this allocation. Our controlled experiments test hypotheses about whether AI enactment improves performance compared to human-only facilitation. The findings will feed into an extended FAM that incorporates facilitation capabilities as first-class constructs with measurable effects (Dennis, Wixom, & Vandenberg, 2001).

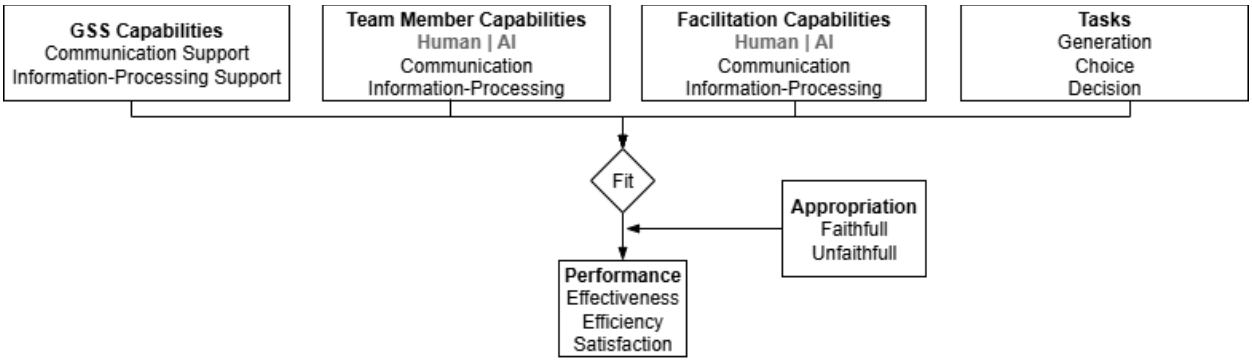

Figure 1: Conceptual model of hybrid facilitation

To operationalise facilitation within FAM, we distinguish two complementary capability classes, communication support and information-processing support, and treat each as a designable bundle that can be enacted by humans, delegated to an AI co-facilitator, or orchestrated in hybrid form as paved by the Collaboration Engineering/ThinkLet lineage of codified, reusable facilitation techniques (Briggs, de Vreede, & Nunamaker, 2003; Kolfschoten, de Vreede, Briggs, & Sol, 2006; Schwabe et al., 2025).

Communication support governs how contributions are exchanged so that interaction remains coordinated, safe, and phase-appropriate (convey vs. converge) in line with Media Synchronicity Theory (Dennis & Valacich, 1999). Concretely, it equips groups with (i) participation balancing and floor control such as through continuous sensing of silence, dominance, overlap and respectful prompting to widen the floor, (ii) pacing and phase governance such as through timeboxing, agenda visibility, and smooth transitions between diverging and converging, (iii) tone and relational-climate regulation such as through para-verbal modulation (empathy, assertiveness, civility cues) and de-escalation nudges, (iv) anonymity orchestration such as through context-specific toggles (e.g., idea generation, critique, voting) with explicit norm signalling, and (v) group-memory cues for communication such as through lightweight capture and resurfacing of "who has spoken/what is pending." These are classic GSS/CE structures that reduce losses like domination and evaluation apprehension while supporting predictable collaboration patterns (Nunamaker et al., 1991; Briggs, de Vreede, & Nunamaker, 2003; Kolfschoten et al., 2006).

We will test whether and with what effects these capabilities can be delegated to an AI co-facilitator that monitors multimodal traces (verbal, para-verbal, selected non-verbal features) and acts through speech, chat, or visual signals. The targeted mechanisms are reductions in process losses such as production blocking, airtime fragmentation, domination, agenda drift, convergence fatigue, evaluation apprehension, attention/recall failures and incivility, alongside increases in process gains such as stimulation, learning, synergy, objective evaluation, and error catching (Nunamaker et al., 1991; Dennis, Wixom, & Vandenberg, 2001).

Primary outcomes comprise efficiency (e.g., time-to-decision, overrun frequency), effectiveness (e.g., equality of voice as a precursor to decision quality), and satisfaction (e.g., perceived fairness, trust, clarity). Exemplar measurements include speaking-turn entropy/Gini indices, interruption rates, adherence-to-plan, and affect/trust scales, in line with established GSS and team-process measurement traditions (Dennis, Wixom, & Vandenberg, 2001).

Information-processing support governs what happens to contributions once expressed, helping groups move from raw input to shared understanding and high-quality decisions. It provides (i) rolling summarisation that reduces overload and surfaces novelty vs. redundancy (Converge/Reduce patterns), (ii) clustering and structuring of ideas into coherent themes and visual maps, highlighting gaps that require additional inquiry (Organize pattern), (iii) consensus checking and conflict surfacing through real-time polling, patterning of agreement/disagreement, and decision matrices (Build Consensus pattern), (iv) evaluation and aggregation via criteria-based rating, ranking, and calibrated voting to temper unequal influence and idiosyncratic bias (Evaluate pattern), and (v) organisational memory and reprocessability by means of persistent, searchable capture of contributions and decisions to minimise rework and accountability gaps (Briggs, de Vreede, & Nunamaker, 2003; Kolfschoten et al., 2006; Nunamaker et al., 1991).

We will test whether and with what effects cognitively demanding analytic operations in this class can be delegated to an AI co-facilitator that performs semantic analysis, aggregation, and consistency checks at speed, while reserving contextual

framing, value trade-offs, and fairness judgements for human facilitators when appropriate. This is a complementary-strengths division of labour consistent with hybrid intelligence and human-AI teaming (Dellermann et al., 2019; Seeber et al., 2020). The targeted mechanisms here are reductions in information overload, fragmentation, incomplete task analysis, hidden profiles, premature convergence, and failure to remember, with concomitant gains in more information, learning, objective evaluation, and catching errors (Nunamaker et al., 1991).

Outcomes and precursors include decision quality, semantic diversity, convergence speed, rework reduction, confidence/legitimacy, and satisfaction with process clarity (e.g., measured via dispersion/novelty indices, quality ratings, variance reduction in preferences, recall accuracy), aligning with canonical effectiveness/efficiency/satisfaction outcome families (Dennis, Wixom, & Vandenberg, 2001).

Across both classes, our delegation logic is principled: AI is expected to excel at continuous monitoring, precise timing, and high-volume analysis; humans excel at legitimacy, norm sensitivity, and contextual judgement. Hybrid orchestration in terms of an AI proposing/monitoring while humans validate and enact delicate interventions is most appropriate under equivocality, power asymmetry, or value-laden trade-offs, reflecting complementary team performance (Dellermann et al., 2019; Seeber et al., 2020) and Collaboration Engineering's emphasis on predictable, appropriable patterns (Briggs, de Vreede, & Nunamaker, 2003). Furthermore, we will embed safeguards (transparent intervention logs, explainable prompts, and human override) to preserve agency and trust, following Coactive Design's OPD principles of Observability, Predictability, and Directability for effective human-machine teamwork (Bradshaw et al., 2012; Bradshaw et al., 2013). The following Table 1 summarises each capability with its AI co-facilitation functions, the primary outcomes it targets, and the underlying process gains/losses.

Capability Type	***Typical Human Facilitator Capabilities***	***AI Co-Facilitation Capabilities***	***Primary Performance Outcomes***	***Rationale from Process Gains and Losses***
Participation Balancing & Floor Control	Balance participation, manage speaking order	Detect silence/dominance, prompt contributions, flag interruptions	Equality of voice ↑	↓ Production blocking & domination
Pacing & Phase Governance	Structure agenda, signal transitions, manage timing	Provide timers, progress displays, adaptive pacing	Time-to-decision ↓	↓ Coordination loss, agenda drift
Tone & Relational Climate Regulation	Maintain civility, reframe tension, show empathy/ firmness, de-escalate conflicts	Detect tension markers, suggest reframings, civility prompts, flag toxic language	Trust ↑	↓ Incivility & defensiveness
Blind-Spot Detection (Coverage of Issue Space)	Monitor breadth of discussion, surface neglected topics	Semantic coverage analysis, highlight gaps, propose probes/contributions	Coverage of issue space ↑ Decision quality ↑	↓ Incomplete analysis, hidden profiles
Clustering & Structuring of Contributions	Group and label ideas, visualise relations, highlight gaps	Semantic clustering, generate visual maps, suggest merges/splits	Shared understanding ↑	↓ Overload & fragmentation
Convergence Support	Summarise progress, highlight (dis)-	Rolling summaries, agreement visualisation,	Convergence speed ↑ Legitimacy of outcome ↑	↓ Premature convergence, hidden disagreement

(Summarisation & Evaluation)	agreements, consensus	test	structured polls & ranking

Table 1: Facilitation capability taxonomy: six core capabilities with facilitator behaviours, AI co-facilitation functions, primary outcomes, and aligned process gains/losses. All capabilities operate over spoken and written streams; participation and turns are computed across channels.

These capability classes can manifest at different structuring levels. At the meeting level, communication support may take the form of agenda pacing or transparent governance, while information-processing support may involve logging decisions. At the activity level, communication support may balance contributions within a brainstorming or consensus round, while information-processing support may cluster or structure ideas. At the micro-process level, communication support may nudge a silent participant or regulate tone, while information-processing support may highlight semantic novelty in real time. This layered view follows the three-level model of structuring tactics in the GSS/CE literature (Niederman, Briggs, de Vreede, & Kolfschoten, 2008).

This layered view shows that the two capability classes are not bound to a single level of intervention; rather, they cut across levels. Our project leverages this taxonomy to design digital artefacts that instantiate specific capabilities, as designable bundles in the ThinkLet tradition, each tied to a hypothesis about how their allocation between humans and digital artefacts influences group performance (Briggs, de Vreede, & Nunamaker, 2003; Kolfschoten et al., 2006; Schwabe et al., 2025). In design science terms, these capabilities and allocation rules form the building blocks of a design theory (Gregor & Jones, 2007). By systematically articulating design principles that specify mechanisms, actors, and rationales (Gregor et al., 2020), we lay the groundwork for a coherent theory of multimodal co-facilitation.

3.2 Digital Artefacts to be Engineered

To instantiate the capability classes and evaluate delegation effects, we will implement six multimodal artefacts. Each observes verbal and para-verbal traces (and selected non-verbal features with consent) and intervenes via voice, chat, or visual cues. In written phases (e.g., brainstorming, polls), artefacts sense message cadence, entry counts/lengths, edit/retract events, and up-votes, and can prompt text contributions or surface under-represented perspectives. A shared group-memory layer logs interventions with rationales, provides live recaps, and preserves decision provenance. All artefacts expose human override.

Airtime Balancer (Communication - Participation balancing & floor control)

The Airtime Balancer supports participation by monitoring basic diarisation and talk-time shares, overlap and interruption markers, and simple turn counts. Based on these signals it offers subtle "next up" cues, optional speaking-order prompts, and short nudges to widen the floor. Its primary outcome is improved airtime equality, with interruption rate, perceived fairness, and satisfaction as secondary indicators.

Process Pacer (Communication - Pacing & phase governance)

The Process Pacer addresses pacing and phase governance by sensing elapsed time against the agenda, response latency, and participation dispersion. It intervenes through visible timers, progress beacons, and phase or goal cards that guide smooth transitions. Its primary outcome is reduced time-to-decision, with process clarity, satisfaction, and overrun frequency as secondary measures.

Tone & Climate Regulator (Communication - Tone & relational climate)

The Tone & Climate Regulator sustains constructive interaction by detecting lightweight tension indicators such as sharp prosodic shifts or negative lexical cues. It responds through private civility prompts, reframing suggestions, and short

norm reminders. The primary outcome is higher trust, with incidence of negative affect, breakdowns avoided, and satisfaction serving as secondary outcomes.

Blind-Spot Detector (Information-Processing - Coverage of issue space)

The Blind-Spot Detector improves coverage by comparing ongoing discussion with a compact issue map seeded from task materials. When gaps appear, it surfaces "missing topics" cards, probe questions, or draft contributions for facilitator review. Its primary outcome is broader issue-space coverage, with decision quality as a co-primary and satisfaction with thoroughness as a secondary indicator.

Structurer (Information-processing - Clustering & structuring of contributions)

The Structurer enhances shared understanding by analysing semantic similarity among contributions and identifying redundancy or novelty. It produces initial cluster maps, proposes renames, merges or splits, and highlights "what's missing." Its primary outcome is shared understanding, supported by decision time, perceived clarity, and rework rate as secondary outcomes.

Convergence Supporter (Information-processing - Summarisation & evaluation)

The Convergence Supporter accelerates decision-making by sensing agreement and disagreement patterns, option stability, and unresolved issues. It intervenes with rolling summaries, agreement heatmaps, one-click polls, and criteria-based rankings with variance flags. Its primary outcome is faster convergence speed, complemented by legitimacy of outcome as a co-primary and by decision quality, confidence, and satisfaction as secondary outcomes.

Architecture & safeguards

All artefacts share a common architecture designed to ensure transparency, reproducibility, and human agency. Interventions are logged with rationales, surfaced to participants through clear handoff cues, and can be overridden immediately by the human facilitator. A shared group-memory layer provides live recap cards ("who has spoken/what's pending"), searchable records, and a decision provenance dashboard, ensuring process transparency and accountability. To guarantee consistency across conditions, language model outputs are generated with deterministic decoding (temperature = 0, nucleus = 0), response caching for standard prompts, and guardrails to prevent content drift. Data governance follows a strict minimisation principle: only task-relevant features are captured, raw audio/video are processed ephemerally, and facial affect, gaze, or posture cues are analysed on-device with immediate disposal of raw frames. These safeguards are complemented by procedural controls in the experiments: facilitators follow a scripted protocol with no discretionary interventions beyond safety, are blinded to conditions except where unavoidable (e.g., visibility of timers), and are modelled either as fixed effects or rotated with random intercepts. Together, these technical and procedural safeguards ensure that co-facilitation artefacts remain accountable, reproducible, and human-centred.

3.3 Experimental Designs

We will systematically test six hypotheses (H1-H6) in controlled team experiments. The experimental unit is the team, composed of four human participants and two AI team members in fixed non-facilitator roles. AI team members contribute content in the task (e.g., ideas, rationales) but do not perform facilitation functions. These use fixed prompts and deterministic decoding to ensure their interventions are bound to artefact interventions. A trained human facilitator guides the team. Facilitation interventions are restricted to the focal capability. A trained human facilitator runs the protocol (instructions, timing cues, ethics/safety) but does not perform additional co-facilitation.

Each team works through a structured "generate → organise → prioritise" task (e.g., idea generation, clustering, and criteria-based decision), which mirrors the well-established "Generate & Choose" task family in the GSS tradition (Dennis et al., 1988; Dennis, Wixom, & Vandenberg, 2001). This task family reliably elicits both communication and information-processing challenges. These tasks mirror real-world collaboration practices while remaining controlled for valid experimentation. For each digital artefact, we run a dedicated experiment comparing conditions where the co-facilitation capability is Off (baseline) vs. On (AI-supported). This design allows us to test directional hypotheses of the form: delegating capability X to AI co-facilitation improves outcome Y (e.g., participation equality, convergence speed, decision quality) relative to human-only facilitation.

We distinguish two stages of evaluation. In formative pretests, we run small sessions with two functions: (1) technical and functional trials of prototypes to tune sensing thresholds, phrasing, and trigger logic, and (2) tests of the fit with other components of the Fit-Appropriation Model (e.g., GSS affordances, team composition, task equivocality). In the main experiments, we then run preregistered, sufficiently powered studies with stable tasks and conditions to provide causal evidence for H1-H6. Primary outcomes include equality of voice, semantic diversity, convergence time, decision quality, and satisfaction/trust; secondary outcomes cover efficiency, clarity, and legitimacy. Manipulation checks capture perceived presence, assertiveness, and transparency of artefact interventions. This design ensures both internal validity (controlled causal tests) and theoretical validity (integration with FAM as appropriation support).

Experiment 1: Airtime Balancer

H1: AI-supported participation balancing reduces airtime inequality and increases fairness vs. human-only facilitation.

Teams of four humans plus two AI teammates perform an idea-generation and discussion task. In the baseline condition, no balancing support is active. In the AI-supported condition, the artefact detects silence, dominance, and overlaps using VAD/SCD/diarisation, prosody, lexical cues, and multimodal sensing (audio + non-verbal intent cues), then suggests turn-taking or issues gentle prompts as well as proposing explicit speaking order. We predict AI-supported participation balancing improves participation equality (lower Gini/entropy of speaking time), with higher perceived fairness and satisfaction and reduced interruption rate. Success criterion: measurable gains in participation balance without declines in satisfaction relative to baseline.

Experiment 2: Process Pacer

H2: AI-supported pacing improves efficiency and clarity without reducing satisfaction.

Teams work through all three phases (generate → organise → prioritise). In the baseline condition, time management is left to participants. The AI-supported condition provides timers, progress beacons, and agenda displays as well as enforcing strict timeboxes and automatically signalling phase transitions, adapted to observed energy/engagement proxies. We predict that AI-supported pacing reduces time-to-decision and overrun frequency, while sustaining or improving perceived process clarity and satisfaction. Success criterion: shorter completion times and fewer overruns without reduced satisfaction compared to baseline.

Experiment 3: Tone & Climate Regulator

H3: AI-supported adaptation sustains trust and reduces breakdowns.

Teams engage in a discussion task with moderate potential for conflict (e.g., prioritising competing proposals). In the baseline condition, no relational climate support is given. In the AI-supported condition, the artefact detects para-verbal tension markers (e.g., pitch spikes, intensity, negative sentiment) and non-verbal cues (facial affect, gaze aversion, laughter dynamics), and provides civility prompts or reframing suggestions. Furthermore, it also flags toxic expressions and issues direct de-escalation cues. We predict higher trust and satisfaction and fewer breakdowns under AI-supported conditions. Success criterion: reduced incidence of negative-affect markers and higher trust scores.

Experiment 4: Blind-Spot Detector

H4: AI-supported blind-spot detection improves coverage of issue space and decision quality.

Teams organise and decide on a set of options. In the baseline condition, no coverage support is provided. In the AI-supported condition, the artefact detects uneven topical focus using semantic analysis and surfaces neglected issues with targeted probes. Furthermore, the artefact also proposes draft contributions that can be adopted or dismissed by the human facilitator. We predict that AI-supported blind-spot detection improves coverage of the issue space and increases decision quality, without overloading participants. Success criterion: broader topical coverage and improved expert-rated decision completeness relative to baseline.

Experiment 5: Structurer

H5: AI-supported clustering/structuring accelerates shared understanding and reduces rework.

Teams cluster and structure ideas generated earlier. In the baseline condition, clustering is manual. In the AI-supported condition, the artefact provides semantic clustering, visual maps, and highlights gaps in the idea set. Additionally, the artefact proposes cluster names and merges, which must be validated by the group. We predict that AI-supported clustering/structuring leads to greater shared understanding and reduced decision time, with participants perceiving higher process clarity. Success criterion: higher consensus on cluster structure and shorter task duration compared to baseline.

Experiment 6: Convergence Supporter

H6: AI-supported convergence support increases convergence speed and legitimacy of outcomes.

Teams move from a set of structured options to a prioritised decision. In the baseline condition, convergence is unassisted. In the AI-supported condition, the artefact generates rolling summaries, highlights agreement/disagreement patterns, and visualises convergence state. Moreover, it also triggers structured polls, criteria-based ranking, and variance/consistency checks. We predict that AI-supported convergence support leads to faster convergence speed and higher perceived legitimacy. Decision quality, confidence, and satisfaction are monitored as secondary outcomes to ensure process gains do not come at the cost of effectiveness. Success criterion: reduced convergence time and higher ratings of legitimacy/confidence in outcomes.

The following Table 2 summarises the six experiments, linking artefacts to their hypotheses, task phases, and primary and secondary outcomes. Each experiment specifies one primary outcome that directly reflects the focal capability and determines sample size and confirmatory tests (e.g., airtime equality for H1, convergence speed for H6). We also pre-specify secondary outcomes that probe mechanisms (e.g., interruption rates) and check for trade-offs (e.g., satisfaction, trust). All outcomes will be preregistered; primaries anchor confirmatory inference, while secondaries provide convergent evidence with multiplicity correction.

Artefact (Hypothesis)	*Task Phase(s)*	*Primary Outcome(s)*	*Secondary Outcomes*
Airtime Balancer *(H1)*	Generate → Discuss	**Airtime equality**	Interruption rate, perceived fairness, satisfaction
Process Pacer *(H2)*	Generate → Organise → Prioritise	**Time-to-decision**	Perceived process clarity, satisfaction, overrun frequency
Tone & Climate Regulator *(H3)*	Discuss → Prioritise	**Trust**	Incidence of negative-affect markers, breakdowns avoided, satisfaction
Blind-Spot Detector *(H4)*	Organise → Prioritise	**Coverage of issue space** **Decision quality**	Shared understanding, satisfaction with thoroughness
Structurer *(H5)*	Organise	**Shared understanding**	Decision time, perceived clarity, rework rate
Convergence Supporter *(H6)*	Prioritise	**Convergence speed** **Legitimacy of outcome**	Decision quality, confidence, satisfaction

Table 2: Experiment Matrix: Artefacts (Hypotheses), Tasks, and Outcomes

Experiments 1-6

We will preregister hypotheses, primary outcomes, exclusion rules, and analysis models (mixed-effects with condition (Off vs. On) as a fixed effect; team and topic as random intercepts). Power analyses (small-medium $f = 0.20$, $\alpha = .05$, $1-\beta = .80$) indicate ≈ 24 teams per condition (exact N in preregistration), yielding robust detection of condition effects in mixed models. Multiplicity for secondary outcomes will be controlled via Holm correction.

We ensure construct validity via a dual measurement approach combining validated psychological scales with objective behavioural telemetry. The integrity of the independent variable (treatment) is assessed by manipulation checks (e.g., perceived presence, assertiveness, transparency, felt agency) and telemetry checks (e.g., prompt rate, adherence to queues/timers). Sessions where checks indicate treatment failure are excluded based on preregistered criteria.

For social-psychological outcomes, we use established, validated instruments: Trust (McAllister, 1995), Perceived Fairness (Colquitt, 2001), Legitimacy and Confidence (Johnson et al., 2014), Satisfaction (Hackman, 1987), and Process Clarity (adapted from Dennis et al., 1996). All survey items (5-7 point Likert) are complemented by objective behavioural/telemetry indicators for robust triangulation. This is critical for effectiveness metrics such as Decision Quality (assessed by expert rubric against criteria; Dean et al., 2006; Phillips, 1984), Coverage of Issue Space (quantified via semantic diversity and completeness scores; Stasser & Titus, 1985), and Shared Understanding (measured through a consensus index and post-session cognitive mapping congruence; Mohammed et al., 2010). Telemetry further captures objective efficiency metrics, including Speaking-Turn Entropy, Interruption Rate, Convergence Time, Time-to-Decision, Overrun Frequency (Nunamaker et al., 1991), Rework Rate, Message-Count Dispersion (text), Text-Turn Gini/Entropy, Entry Latency, and Incidence of Negative-Affect Markers. Together, these validated scales and established measurement traditions ensure both manipulation validity and measurement validity, providing a robust basis for causal inference and theoretical generalisation.

3.4 Work Packages

WP	TITLE	LEAD	CONTRIBUTORS	DURATION	DELIVERABLES
WP1	Project Management	UZH	All	M1-M42	Management reports, ethics approvals and DMP, shared repositories
WP2	Artefact Design & Implementation	ZHAW	PI ZHAW 1 PhD ZHAW	M3-M36	Catalogue of design principles, Open-source artefact library,

			Dev support (ZHAW & IU)		Publications (IS/HCI)
WP3	Experiments: Communication Support	ZHAW	1 PhD ZHAW Dev support (ZHAW & IU)	M1-M42	Public datasets, analysis reports and preregistration packages Publications (CHI/CSCW)
WP4	Experiments: Information-Processing	UZH	1 PhD UZH Dev support (ZHAW & IU)	M1-M42	Public datasets, analysis reports, preregistration packages Publications (HICSS/ICIS/AIS)
WP5	Theory Development	UZH	PI UZH Both PhDs	M1-M42	Extended FAM, synthesis datasets, analysis code Publications (FT50 journal, practitioner outlet)

WP1: Project Management (M1-M42) — Lead: UZH, Contribution ZHAW

Objective: Deliver the project on time, on scope, and compliant with ethics and data protection.

- Governance: bi-weekly PI sync; quarterly steering with decision log; cross-site issue tracker.
- Ethics & data protection: IRB/DPIA approvals; data-minimisation and consent templates enforced.
- Open science: preregistration before data collection; controlled releases of code/data.
- Recruitment & operations: participant panel, scheduling, compensation; risk-gate reviews at M12/M24/M36.

Deliverables: Management reports (meeting minutes, decision logs, risk register), ethics approvals and Data Management Plan (DOI-registered), shared repositories (code, data) with controlled access.

Milestones: M3 Ethics & DMP approved; M12/M24/M36 risk-gates; M42 final report.

WP2: Artefact Design & Implementation (M3-M36) — Lead: ZHAW, Contribution IU, UZH

Objective: Engineer six multimodal co-facilitation artefacts as expository instantiations with safeguards.

- Architecture & principles: define shared sensing/intervention backbone and dose (advisory/assertive) policies.
- Co-design & appropriability: facilitator workshops to tune phrasing, visibility, and override.
- Implementation: capability-specific artefacts with deterministic settings and intervention logs.

Deliverables: Catalogue of design principles, open-source artefact library, publications (IS/HCI).

Milestones: M6 Set A (Airtime Balancer, Blind-Spot Detector); M18 Set B (Process Pacer, Structurer); M24 Tone & Climate Regulator ready (Set C-Part 1); M30 Hardened releases (H1-H4); M36 Convergence Supporter ready (Set C-Part 2) + final library.

WP3: Experiments: Communication Support (M1-M42) — Lead: ZHAW, Contribution UZH

Objective: Causal tests for H1-H3 (participation, pacing, climate).

- Preregistration & power: primary DV per study; mixed-effects models; manipulation checks.
- Pilots → main studies: calibrate thresholds and prompts; run randomised Off vs On conditions.
- Analysis & reporting: confirmatory contrasts; robustness checks; public datasets and code.

Deliverables: Public datasets (de-identified) with codebooks, analysis reports and preregistration packages, conference submissions (CHI/CSCW).

Milestones: Starting with M12, one experiment will be finished every 12 months (i.e. M12, M24, M36)

WP4: Experiments: Information-Processing (M1-M42) — Lead: UZH, Contribution ZHAW

Objective: Causal tests for H4-H6 (blind-spots, structuring, convergence).

- Preregistration & rubrics: decision-quality/coverage rubrics; efficiency/effectiveness telemetry.
- Pilots → main studies: stable artefacts from WP2; randomised Off vs On; full telemetry capture.
- Analysis & reporting: mixed-effects; moderator/mediation; replication packages.

Deliverables: Public datasets (de-identified) with codebooks, analysis reports, and preregistration packages, publications (HICSS/ICIS conferences + AIS premier journals).

Milestones: Starting with M6, one experiment will be finished every 12 months (i.e. M6, M18, M30, M42)

WP5: Theory Development (M1-M42) — Lead: UZH, Contribution IU, ZHAW

Objective: Extend FAM with capability micro-foundations and allocation rules; consolidate design theory.

- Formalisation: embed communication/information-processing supports as appropriation mechanisms.
- Allocation rules: derive human vs AI vs hybrid rules from effect patterns and boundary conditions.
- Synthesis: cross-study meta-summaries; mechanism maps linking micro-metrics to outcomes.
- Translation: practitioner playbook; teaching cases; design-principle catalogue.

Deliverables: Extended Fit-Appropriation Model with capability-level micro-foundations, synthesis datasets and analysis code, publications (FT50 journal + practitioner article).

Milestones: every 6 months a theory workshops alternating between co-located and distributed. Each results in new theory versions and conference and premier IS journal publications.

3.5 Research Team Members' Roles

The project requires the integration of theory-driven group facilitation research (UZH) with the technical design of multimodal AI artefacts (ZHAW), complemented by Indiana University's expertise in digital humans. Each institution supervises a dedicated PhD and contributes senior staff to ensure both timely delivery of artefacts and sustained theoretical contributions.

Prof. Gerhard Schwabe (UZH, Applicant) leads WP1 (Project Management), WP4 (Experiments: Information-Processing), and WP5 (Theory Development). He is responsible for embedding facilitation capabilities into the Fit-Appropriation Model (FAM), specifying allocation rules, and ensuring experimental validity. He supervises both PhDs jointly with Prof. de Spindler and co-leads WP3 experiments, with emphasis on experimental design, micro-process metrics, and theoretical interpretation.

Prof. Alexandre de Spindler (ZHAW, Applicant) leads WP2 (Artefact Design & Implementation) and WP3 (Experiments: Communication Support). He is responsible for the technical architecture of multimodal artefacts using the PROMISE and SBR frameworks, supervises the ZHAW PhD and development staff, and ensures robust sensing, intervention logic, and responsible AI safeguards. He also coordinates technical publications and open-source dissemination.

Postdoc UZH contributes focused expertise on socio-technical aspects of human-AI collaboration. He supports WP3 and WP4 by ensuring measurement validity for micro-process metrics, designing telemetry checks, and co-developing rubrics for decision quality and legitimacy. In WP2 he reviews artefact design for appropriability, transparency, and

override, while in WP5 he integrates empirical findings into FAM extension with particular focus on fairness, trust, and legitimacy. He also coordinates open science practices, codebooks, and de-identification workflows to ensure reproducibility.

UZH PhD Candidate focuses on WP4 and WP5, with embedded collaboration in WP3 to align measurement and analysis with artefact design. He collects and analyses multimodal group data, develops micro-process metrics (airtime equality, convergence speed, semantic diversity), and links empirical findings to FAM extension by specifying capability-level micro-foundations and legitimacy conditions. He contributes to preregistration, meta-analytic synthesis, and co-authors publications in IS, CSCW, and social psychology outlets.

ZHAW PhD Candidate concentrates on WP2 and WP3, while contributing to WP4 to ensure artefacts support rigorous information-processing experiments. He implements artefact prototypes (e.g., Airtime Balancer, Process Pacer, Climate Regulator, Blind-Spot Detector, Structurer, Convergence Supporter) with PROMISE/SBR, ensures reproducible multimodal pipelines and safeguard mechanisms, and abstracts design principles into a design theory of multimodal co-facilitation. He co-authors technical and design-science contributions in HCI, AI, and IS venues.

Development Support (ZHAW) brings more than a decade of software engineering experience from prior applied R&D projects (e.g. Innosuisse). He ensures end-to-end engineering of artefacts, from requirements elicitation and human-centred design to robust prototype implementation and deployment. He maintains code quality, CI/CD pipelines, and containerisation for reproducibility, and supports operational reliability during experiments. His senior expertise ensures the artefact library meets both research and practical standards and remains usable beyond the project.

Prof. Alan Dennis (IU) contributes to WP2 and WP5 through his expertise in digital humans and GSS theory. He participates in regular theory workshops, supports cross-site experiment alignment, and helps integrate findings into a broader theory of hybrid facilitation.

Development Support (IU, 20% FTE) provides focused contributions to WP2 by adapting IU's digital human technologies for integration into the artefact architecture. He develops reusable modules for embodied interaction and dialogue state management, supports cross-site deployment in Zurich and Indiana to ensure parity in experiments, and contributes documentation and reproducible code to the open-source library.

Together, the team combines complementary expertise in theory, artefact design, socio-technical fairness, and robust software engineering. Senior Applicants and the Partner provide strategic leadership, PhDs ensure sustained academic output, and experienced development staff guarantee feasibility and timely implementation. The result is a complete and well-balanced team capable of delivering both rigorous theory and practical artefacts within the proposed timeframe.

3.5 Risks, Mitigations, and Contingencies

We focus on four material risks spanning technical, experimental, validity, and theory. Ethics/compliance and operational risks are handled in WP1 (IRB/DPIA, DMP, governance). Risk gates at M12 / M24 / M36 review indicators and trigger the contingencies below.

Risk	Indicator	Mitigation	Contingency	Owner
Sensing underperforms	Low diarisation accuracy; >10% turn errors	Audio-first pipeline, threshold calibration	Disable non-verbal cues; revert to audio-only	ZHAW PhD
Model/API drift	Output variance ↑ on fixed prompts	Pin model versions; CI evals	Switch to backup/local model; freeze build	ZHAW Dev
Participant reactance	Reactance scale ↑; satisfaction ↓	Private nudges; transparent "why" cards	Reduce assertiveness; route via facilitator	WP3 Lead
Manipulation failure / tiny effects	Checks fail; effect sizes < prereg	Pilot calibration; prereg exclusion rules	Increase equivocality; raise N; refine phrasing	WP3/4 Leads

4 Schedule & Milestones

t	***Milestone***	***Linked WPs***	***Key Deliverables*** *(RX = Round X)*
M6	PM framework operational; Ethics & DMP approved; Architecture & spec freeze; Prototype Set A ready (Airtime Balancer, Blind-Spot Detector), prereg & pilots for Experiment 1	WP1, WP2, WP5	Kick-off & repos; ethics/DMP; prototype Set A
M12	Experiment 1 (H1 Airtime Balancer) completed; prereg & pilots for Experiment 4	WP3	Dataset + analysis R1; CHI/CSCW submission
M18	Prototype Set B ready (Process Pacer, Structurer); Experiment 4 (H4 Blind-Spot Detector) completed, prereg & pilots for Experiment 2	WP2, WP3	Prototype Set B; Dataset + analysis R2
M24	Experiment 2 (H2 Process Pacer) completed; prereg & pilots for Experiment 5; Prototype Set C-Part 1 ready (Tone & Climate Regulator); interim theory memo (WP5)	WP2, WP3, WP5	Dataset + analysis R3; prereg Exp4
M30	Experiment 5 (H5 Structurer) completed; hardened releases of artefacts (H1,H2,H4,H5); prereg & pilots for Experiment 3	WP4	Dataset + analysis R4; HICSS/ICIS submission window
M36	Experiment 3 (H3 Tone & Climate Regulator) completed; Prototype Set C-Part 2 ready (Convergence Supporter); prereg & pilots for Experiment 6; synthesis workshop II	WP2, WP4	Dataset + analysis R5; MISQ/ISR journal submission prep
M42	Experiment 6 (H6 Convergence Supporter) completed; Synthesis & dissemination	WP3, WP4, WP5	Extended FAM theory paper; practitioner article; open artefact library, datasets, design guidelines

5. Relevance and Impact

Scientific Contributions. This project makes the following two complementary contributions. We extend the Fit-Appropriation Model (FAM) with capability-level foundations of facilitation and provide causal evidence from team experiments on when and how AI delegation improves collaboration outcomes (Theory & Evidence). We develop and test six multimodal artefacts as expository instantiations, deriving allocation rules and design principles for human-AI co-facilitation (Design & Principles). Together, these advances yield a mid-range theory of hybrid facilitation and a portable repertoire of design guidelines.

Societal and Economic Relevance. Skilled facilitation is scarce, yet collaboration quality critically shapes outcomes in healthcare, education, public administration, and innovation. Our artefacts and guidelines help teams to (i) run structured discussions with clearer participation and pacing, (ii) move from dispersed input to shared understanding more quickly, and (iii) make decisions with traceable rationales that stakeholders perceive as fair and legitimate. Stakeholders include facilitators, managers, and teams in hospitals, universities, finance, and the public sector. The project augments rather than

replaces human facilitation, enabling wider access to good processes while preserving oversight, in line with SNSF's use-inspired basic research remit.

Responsible AI and Compliance. We embed responsible AI in design and experimentation: interventions logged with rationales; explicit hand-offs; always-on human override (transparency & agency), PROMISE orchestration and deterministic decoding ensure consistent behaviour; prompts are cached and versioned (predictability), and only task-relevant features captured; raw audio/video processed ephemerally; facial/gaze cues analysed on-device; strict de-identification and access control (data protection). These safeguards align with Swiss and EU principles on transparency, human oversight, and data minimisation.

Open Science and Dissemination. We commit to reproducible, open outputs: **Artefact library** with open-source prototypes and experiment harness. **Anonymised datasets** with codebooks and preregistered analysis scripts. **Scholarly venues** in MISQ, ISR, JMIS, CHI, CSCW, HICSS, ICIS. **Practitioner outreach** via facilitator playbooks and short management articles. **Teaching ressources** including cases and lab exercises with telemetry dashboards.

Capacity Building and Collaboration. The project trains two PhD researchers spanning theory (UZH) and multimodal systems (ZHAW). Ongoing collaboration with professional facilitators ensures usability and transfer; cross-site work with IU strengthens generalisability.